

30th International Congress on Sound and Vibration



DISPLACEMENT CALCULATION BY USING PIEZOELECTRIC ACCELEROMETERS

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This study develops a novel approach to obtain displacement information of a unit under test by using piezoelectric accelerometer data taken from random vibration and shock environments. Direct integration of raw random acceleration data causes offset and slope in the velocity and displacement. In this approach, collected acceleration data is represented using series expansion, and integrations are taken analytically; hence, the numerical errors coming from the direct double integration are avoided. Filtering is applied to neglect the frequency range where accelerometers cannot measure correctly. The displacement results obtained using the proposed method are compared with the displacement data measured at the specific locations in the finite element analysis. The proposed method is solid and helpful for calculating the displacement of a specific spot on a unit under test for random vibration and shock tests when a displacement sensor is not applicable due to physical limitations.

Keywords: displacement from accelerometer, random vibration, digital signal processing

1. Introduction

Vibration tests provide a wealth of information about the lifespan and durability of a component or system under different operating conditions. Generally, acceleration, displacement, and stress or strain values are measured in these tests, with piezoelectric accelerometers, laser displacement sensors, strain gauges are commonly used. Despite the apparent derivative or integral relationship between the measured parameters, it is preferred to measure each of them separately with unique sensors. This is because the transformations between them, whether due to noise, mathematical losses during the transition from analog to digital, or the capacities of the sensors, may not yield the desired quality. In this study, the method of measuring relative displacement between two spots on a system with two piezoelectric accelerometers has been investigated.

This topic has been studied both at the industry and academic levels. The Siemens LMS team [1] suggested extracting a sixth-degree polynomial from accelerometer data to prevent non-zero averaging and accumulation of errors over time before integrating, which is a method developed through experience. The Vibration Research team [2] worked on an algorithm that operates only on sinusoidal waves with known frequencies. The Wilcoxon team [3] is commercializing their own developed system in this regard. Freescale team [4] developed a procedure relying on the moving averages to double integrate acceleration data to obtain displacement. Ferrero et al. [5] worked on noise and measurement errors by incorporating accelerometer mechanics with a Kalman filter and estimator. Ribeiro et al. [6] focused on the sinusoidal vibration response in systems with small amounts of average and error drift, and they conducted experiments with different kinds of accelerometers to match their findings with the real results. Abir et al. [7] modified the collected data with a lead-lag compensator and two frequency domain filters, optimizing the parameters in their models with a genetic algorithm. Han [8] worked on the analog-to-digital conversions of numerically generated single-degree-of-freedom harmonic signals. Different algorithms have also been developed in cases where the modal parameters of the system being measured are known [9, 10].

In this study, an analytical approach has been developed to calculate the relative displacement between two spots on a system under random vibration and shock inputs. For this method to operate with an acceptable error, the measured system's initial natural frequency must be at least 50 Hertz.

2. Theory

The data collected from an accelerometer can be represented analytically as follows:

$$A = a_0 + \sum_{n=1}^{m/2} a_{s_n} \cdot \sin(n\omega_0 t) + \sum_{n=1}^{m/2} a_{c_n} \cdot \cos(n\omega_0 t), \quad 0 \leq t \leq t_1. \quad (1)$$

In the above equation, the values m and ω_0 are directly related to the data acquisition system, and t_1 represents the time data collected. The accelerometer position at the start and the end of a vibration experiment should be same; hence, the mean of the data should be zero, assuming that there is no plastic deformation. The equation becomes:

$$A = \sum_{n=1}^{m/2} a_{s_n} \cdot \sin(n\omega_0 t) + \sum_{n=1}^{m/2} a_{c_n} \cdot \cos(n\omega_0 t), \quad 0 \leq t \leq t_1. \quad (2)$$

If we represent the data as above, one can directly double integrate analytically to obtain the displacement:

$$A = -\sum_{n=1}^{m/2} \frac{a_{s_n}}{(n\omega_0)^2} \cdot \sin(n\omega_0 t) - \sum_{n=1}^{m/2} \frac{a_{c_n}}{(n\omega_0)^2} \cdot \cos(n\omega_0 t), \quad 0 \leq t \leq t_1. \quad (3)$$

Yet, to calculate the position, the coefficients of harmonic functions are necessary. An intermediate step can be created to use a unitless time as follows:

$$\theta = \frac{t}{t_1} \cdot 2\pi, \quad 0 \leq \theta \leq 2\pi. \quad (4)$$

Then, the Eq. 2 becomes:

$$A = \sum_{n=1}^{m/2} a_{s_n} \cdot \sin\left(n\omega_0 \frac{t_1\theta}{2\pi}\right) + \sum_{n=1}^{m/2} a_{c_n} \cdot \cos\left(n\omega_0 \frac{t_1\theta}{2\pi}\right), \quad 0 \leq \theta \leq 2\pi. \quad (5)$$

In the above equations, the unit of the frequency, ω_0 , is rad/s. To change it to Hertz:

$$f_0 = \frac{\omega_0}{2\pi}, \quad (6)$$

$$A = \sum_{n=1}^{m/2} a_{s_n} \cdot \sin(nf_0 t_1 \theta) + \sum_{n=1}^{m/2} a_{c_n} \cdot \cos(nf_0 t_1 \theta), \quad 0 \leq \theta \leq 2\pi. \quad (7)$$

Using the orthogonality of the harmonic functions, the coefficients of the harmonics can be calculated with the following integrals:

$$a_{s_n} = \frac{1}{\pi} \int_0^{2\pi} A \sin(nf_0 t_1 \theta) d\theta, \quad n = 1, 2, \dots, m \quad (8)$$

$$a_{c_n} = \frac{1}{\pi} \int_0^{2\pi} A \cos(nf_0 t_1 \theta) d\theta, \quad n = 1, 2, \dots, m \quad (9)$$

In the above equations, A represents the data collected, the undefined parameter t_1 is the last second of the recording, and the f_0 value can be calculated as follows:

$$f_0[\text{Hz}] = \frac{f_n}{m} \cdot \frac{\text{sampling frequency} [\text{Hz}]}{\text{length of data}} \quad (10)$$

The ω_0 value can be obtained through the Eq. 6.

2.1 Critical points while conducting the experiment

For the Fourier Transform to yield accurate results, it is crucial for the data to be periodic. Therefore, initiating the recording of time data before the test begins and concluding the recording of time data after the test is entirely finished is essential. This can be easily accomplished in both random vibration and shock tests. Secondly, due to sensor noise, the usage 0.02 Tukey window is strongly recommended to ensure the periodicity.

2.2 Critical points in the frequency domain

Most of the piezoelectric accelerometers are not capable of measuring the vibration below 1.5 Hz. Secondly, due to some digitalization errors, the rigid body motion, which occurs at 0 Hz and measured with inertial measurement units, may have components on the low frequencies. When expressing position data analytically, low frequencies where accelerometers cannot measure should be excluded. It is suggested to omit frequencies below 5 Hz.

2.3 The systems where the method can be applied

Recognition of the system's rigidity is essential in frequency ranges where the accelerometers utilized lack accuracy in measurements. Otherwise, the relative displacements cannot be measured via accelerometers and the results will have significant amount of error.

For a system with one degrees of freedom with structural damping, the general response in frequency domain is as follows:

$$\frac{X}{F} = \frac{1}{k + ih - m\omega^2} \quad (11)$$

Where, ω represents the frequency of the harmonic input, k is for the stiffness of the system, m is the mass of the system, and h represents the structural damping of the system. The same equation can also be written as follows:

$$\frac{X}{F} = \frac{1/k}{1 + i\gamma - \frac{\omega}{\omega_n}} \quad (12)$$

In the above equation, γ represents the loss factor, which is the ratio between structural damping and stiffness of the system, and ω_n stands for natural frequency of the system. In general, the value of the loss factor can be taken between 0.03 and 0.1; in fact, the default value of the loss factor is 0.05 in many different finite element softwares. If the accelerometers used are not capable of making accurate measurement below ω_a [Hz], the transmissibility at ω_a becomes:

$$Tr(\omega_a) = \frac{1}{1 + i\gamma - \left(\frac{\omega_a}{\omega_n}\right)^2}. \quad (13)$$

Rigid body motion means the transmissibility to be 1. If 1% difference between the whole motion and rigid body motion of the system is acceptable, and with a low loss factor as 0.03, the transmissibility equation becomes:

$$1.01 = \frac{1}{\left|1 + i \cdot 0.03 - \left(\frac{\omega_a}{\omega_n}\right)^2\right|}. \quad (14)$$

Above equation leads the first natural frequency of the system has to be at least tenfold of the ω_a . In the Fig. 1 transmissibility of a system as a function of ω_a/ω_n is depicted.

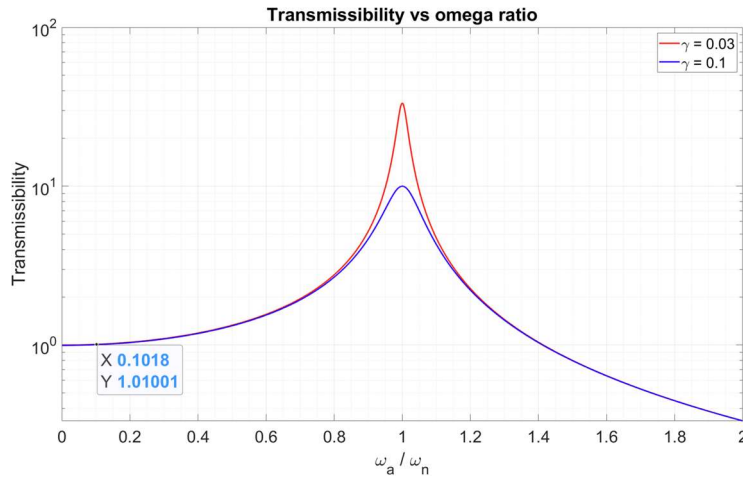


Figure 1: Effect of different gamma values to the transmissibility.

3. Application

To prove the proposed approach, finite element analysis results will be presented. Four beams with different lengths are modelled in Abaqus 2022, as depicted in Fig. 2. Random acceleration input is given to one end of the beams for two seconds, and acceleration and displacement outputs are requested at both ends, at 1 kHz sampling frequency, in the explicit dynamic solution.

The first natural frequency of a fixed-free beam can be calculated as follows [11]:

$$\omega_n = \left(\frac{1.875}{L}\right)^2 \sqrt{\frac{EI}{\rho A}}. \quad (15)$$

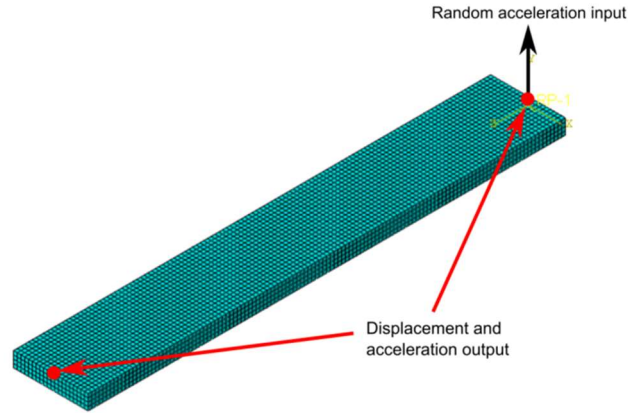


Figure 2: Finite element model with C3D8R elements.

Parameters used in the modelling of different beams in the finite element analysis are presented in Table 1.

Table 1: Parameters of the beams.

Beam	E	b	h	ρ	L	ω_n
2	200 GPa	50 mm	10 mm	7800 kg/m ³	320 mm	79.87 Hz
3	200 GPa	50 mm	10 mm	7800 kg/m ³	400 mm	51.12 Hz
4	200 GPa	50 mm	10 mm	7800 kg/m ³	570 mm	25.17 Hz
5	200 GPa	50 mm	10 mm	7800 kg/m ³	900 mm	10.1 Hz

The acceleration outputs taken from the finite element model will also contain information at the frequencies where piezoelectric accelerometers cannot operate properly. The contributions of different frequency ranges can be separated as follows:

$$a_{in} = a_{in0-\omega_a} + a_{in\omega_a-500} \quad (16)$$

$$a_{out} = a_{out0-\omega_a} + a_{out\omega_a-500} \quad (17)$$

By taking the ω_a value as 5 Hz, the relative acceleration between the input and the output can be written as follows:

$$a_{rel} = (a_{out0-\omega_a} - a_{in0-\omega_a}) + (a_{out\omega_a-500} - a_{in\omega_a-500}). \quad (18)$$

If it is assumed that the system will act rigidly below 5 Hz, i.e. its first natural frequency is bigger than 50 Hz, the relative acceleration can be represented as follows:

$$a_{rel} = a_{out5-500} - a_{in5-500} = A. \quad (19)$$

By using Eqs. 1-10, the coefficients of the harmonic terms can be calculated and the required integrals can be taken analytically. To ensure the periodicity of the collected data, windowing is necessary. 0.02 Tukey window is used, and the relative displacement results are depicted in Fig. 3.

The proposed approach gives more accurate results as the first natural frequency of the system gets higher. The main reason the assumption that the system will act rigidly below 5 Hz. To observe the errors in the results, Fig 4. represents a narrower time span to have a closer look.

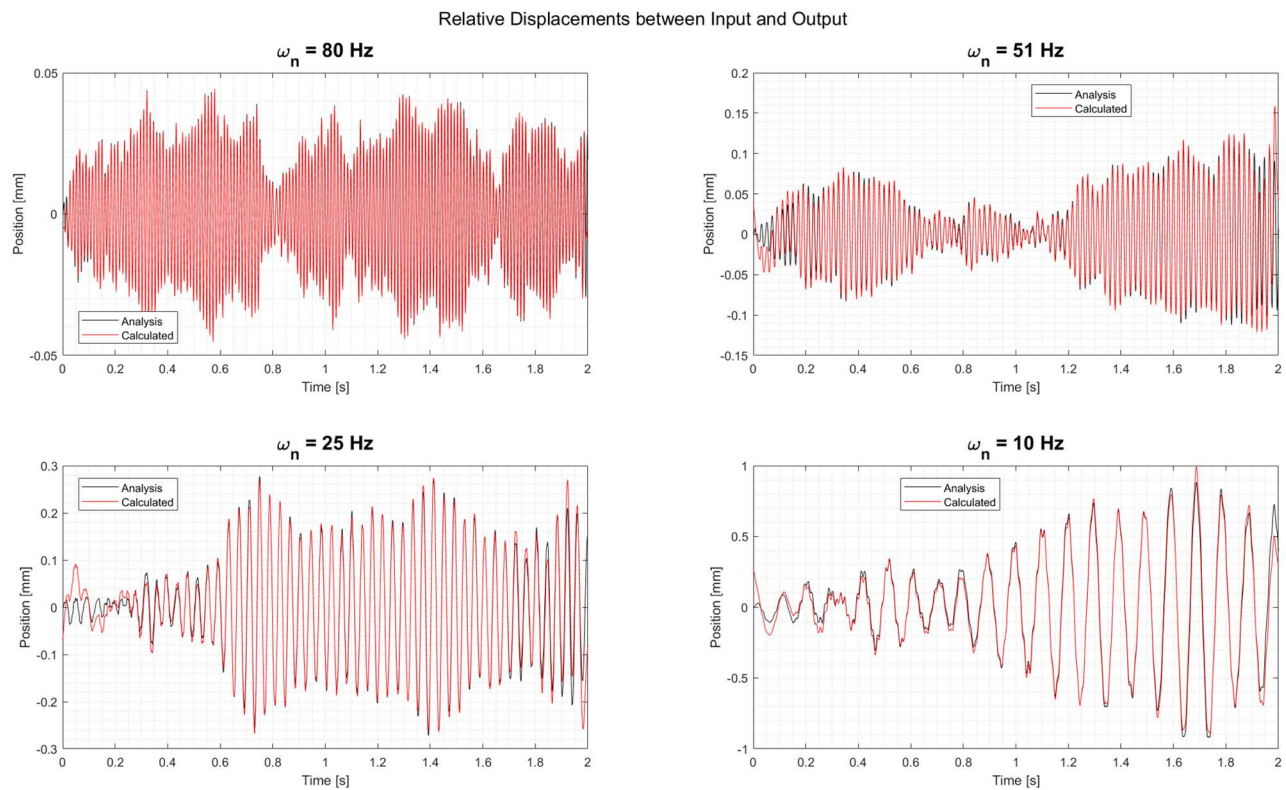


Figure 3: Relative displacements.

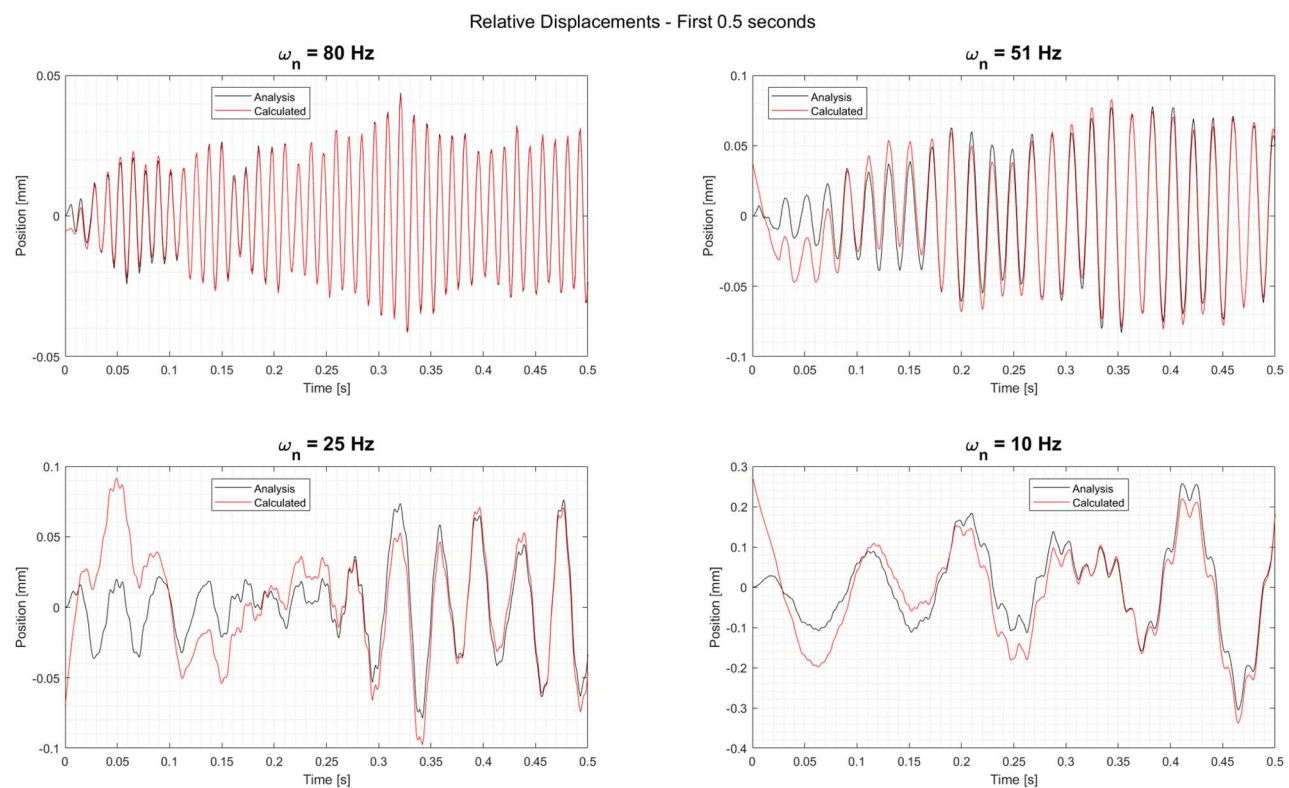


Figure 4: Relative displacements at the first half second.

In this study, the chosen windowing plays a crucial role. Since there will be repeated transitions between frequency and time domains, magnitudes are vital alongside frequency information. While ensuring the periodicity of the collected signal, it is inevitable to also affect its magnitude. However, this effect can be minimized with a small coefficient tapered cosine window. Therefore, a 0.02 Tukey window has been used.

This method can only be used to calculate the relative displacement difference between two accelerometers. The error in the results are directly related to the first natural frequency of the system, and the measurement capacities of the used accelerometers.

4. Conclusion

This paper presents an approach to calculate relative displacement between two spots using two piezoelectric accelerometers. This approach has been validated using finite element analysis, and the relative displacement between two spots obtained with the method and directly from the analysis are compared. The error between the analysis results and the proposed method mainly related to first natural frequency of the system of interest as indicated in the theory section.

This method can be used in real time applications easily. It is required to collect time data from the shaker and the placed accelerometers to calculate the relative displacement between the ground and the spot. Time data collection should be started before the vibration experiment and should be stopped after the test is conducted to ensure starting and ending points of the collected data to be zero, or only have minimal noise. Then, again to strengthen the periodicity of the data, usage of 0.02 Tukey window is suggested. The correct functioning of the recommended method relies not on which test the gathered data is obtained from, but rather on the first natural frequency of the unit under test. Although the details are not explained within this article, the recommended method has also been tested in both random and shock tests and has yielded successful results.

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